



Cambridge International AS & A Level

PHYSICS

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Paper 4 A Level Structured Questions

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MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Annotations

✓	Indicates the point at which a mark has been awarded.
X	Indicates an incorrect answer or a point at which a decision is made not to award a mark.
XP	Indicates a physically incorrect equation ('incorrect physics'). No credit is given for substitution, or subsequent arithmetic, in a physically incorrect equation.

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ECF	Indicates 'error carried forward'. Answers to later numerical questions can always be awarded up to full credit provided they are consistent with earlier incorrect answers. <u>Within</u> a section of a numerical question, ECF can be given after AE, TE and POT errors, but not after XP.
AE	Indicates an arithmetic error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
POT	Indicates a power of ten error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
TE	Indicates incorrect transcription of the correct data from the question, a graph, data sheet or a previous answer. For example, the value of 1.6×10^{-19} has been written down as 6.1×10^{-19} or 1.6×10^{19} . Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
SF	Indicates that the correct answer is seen in the working but the final answer is incorrect as it is expressed to too few significant figures.
BOD	Indicates that a mark is awarded where the candidate provides an answer that is not totally satisfactory, but the examiner feels that sufficient work has been done ('benefit of doubt').
CON	Indicates that a response is contradictory.
I	Indicates parts of a response that have been seen but disregarded as irrelevant.
M0	Indicates where an A category mark has not been awarded due to the M category mark upon which it depends not having previously been awarded.
^	Indicates where more is needed for a mark to be awarded (what is written is not wrong, but not enough). May also be used to annotate a response space that has been left completely blank.
SEEN	Indicates that a page has been seen.

Question	Answer	Marks
1(a)	acceleration perpendicular to velocity	B1
1(b)(i)	decreases	B1
1(b)(ii)	(acceleration of) 9.8 m s^{-2} is caused by weight of car or centripetal force must be greater than weight of car	B1
	(acceleration $> 9.8 \text{ m s}^{-2}$) requires contact <u>force</u> from track or (centripetal force $>$ weight) requires contact <u>force</u> from track	B1
1(c)	$\frac{1}{2}mv_Y^2 = \frac{1}{2}mv_X^2 - mgh$	C1
	$a = v^2 / r$	C1
	$v_Y^2 = 3.8^2 - 2 \times 9.81 \times 0.62$ so $v_Y = 1.5 \text{ m s}^{-1}$ $a = 1.5^2 / 0.31 = 7.3 \text{ m s}^{-2}$ (which is less than 9.8 m s^{-2}) so no	A1
	or	
	$v_Y = \sqrt{(9.81 \times 0.31)} = 1.74 \text{ m s}^{-1}$ so $v_X^2 = 1.74^2 + 2 \times 9.81 \times 0.62$ $v_X = 3.9 \text{ m s}^{-1}$ (which is greater than 3.8 m s^{-1}) so no	(A1)
1(d)	acceleration is independent of mass so makes no difference or mass cancels in the equation so makes no difference	B1

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Question	Answer	Marks
2(a)	(gravitational) field strength equals (gravitational) potential gradient	M1
	reference to minus sign	A1
2(b)(i)	potential is zero at infinity	B1
	(gravitational) force is attractive	B1
	(test) mass getting closer (from infinity) loses potential energy	B1
2(b)(ii)	- potential at (surface of) planet is smaller than at (surface of) moon - potential gradient at (surface of) planet is smaller than at (surface of) moon - magnitude of potential varies inversely with distance from centre near the spheres (point of) maximum potential is nearer to moon than planet <i>Any two points, 1 mark each</i>	B2
2(b)(iii)	sketch: one curve, starting with gradient of decreasing magnitude at $2R$ and finishing with gradient of increasing magnitude at $D - R$	B1
	field strength shown as zero (only) near the point of maximum potential	B1
	negative field strength near one sphere and positive field strength near the other	B1

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Question	Answer	Marks
3(a)(i)	no loss of <u>kinetic</u> energy	B1
3(a)(ii)	• <u>molecules</u> have negligible volume (compared with gas/container) • no forces between <u>molecules</u> (except during collisions) • <u>molecules</u> are in random motion • collisions are instantaneous Any two points, 1 mark each	B2
3(b)(i)	$2mu$	A1
3(b)(ii)	$2L / u$	A1
3(b)(iii)	force = change in momentum / time = $2mu / (2L / u)$ $= mu^2 / L$	A1
3(b)(iv)	pressure = force / area = $(mu^2 / L) / L^2$ $= mu^2 / L^3$	A1
3(c)	$pV = NkT$	C1
	$NkT = \frac{1}{3}Nm\langle c^2 \rangle$ leading to $\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$ <u>and</u> $\frac{1}{2}m\langle c^2 \rangle = E_K$	A1
3(d)	$\frac{1}{2} \times 3.34 \times 10^{-27} \times \langle c^2 \rangle = (3/2) \times 1.38 \times 10^{-23} \times (25 + 273)$	C1
	r.m.s. speed = $1.9 \times 10^3 \text{ m s}^{-1}$	A1

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Question	Answer	Marks
4(a)	straight line through the origin	B1
	negative gradient	B1
4(b)	$a = (-)\omega^2 x$ and $T = 2\pi / \omega$	C1
	e.g. $\omega = \sqrt{(0.80 / 0.12)}$ (<i>any correct pair of values of a and x</i>) $(= 2.58 \text{ rad s}^{-1})$	C1
	$T = 2\pi / 2.58$ $= 2.4 \text{ s}$	A1
4(c)(i)	Point labelled P at one end of the line	B1
4(c)(ii)	Point labelled Q at displacement with magnitude more than half but less than maximum	B1

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Question	Answer	Marks
5(a)(i)	unmodulated (radio) waves would interfere with each other or not modulating would require aerials too long (to be practical)	B1
5(a)(ii)	advantage: • can transmit higher frequencies • higher quality <u>reproduction</u> • less prone to interference • same frequency can be used in different areas (any one point)	B1
	disadvantage: • takes up greater bandwidth • shorter range of transmission • requires a greater number of transmitting aerials (any one point)	B1
5(b)	AM amplitude: min. 8 mV and max. 12 mV	B1
	AM frequency: min. 100 kHz and max. 100 kHz	B1
	FM amplitude: min. 10 mV and max. 10 mV	B1
	FM frequency: min. 90 kHz and max. 110 kHz	B1
5(c)	8.4 kHz	A1

Question	Answer	Marks
6(a)	work done per unit charge	B1
	(work done in) moving positive charge from infinity	B1
6(b)	$C = Q / V$	C1
	$V = Q / (4\pi\epsilon_0 r)$ and so $C = Q / [Q / (4\pi\epsilon_0 r)] = 4\pi\epsilon_0 r$	A1
6(c)	$Q = 4\pi\epsilon_0 r V = 4\pi \times 8.85 \times 10^{-12} \times 0.13 \times 4500$ $(= 6.5 \times 10^{-8} \text{ C})$	C1
	$(Q - q) / 13 = q / 5.2$	C1
	$5.2Q - 5.2q = 13q$, so $q = (5.2 / 18.2)Q$ $q = (5.2 / 18.2) \times 6.5 \times 10^{-8}$ $= 1.9 \times 10^{-8} \text{ C}$	A1
	or	
	$V_T = Q_T / C_T$ $= 6.5 \times 10^{-8} / [4\pi \times 8.85 \times 10^{-12} \times (0.13 + 0.052)]$ $(= 3210 \text{ V})$	(C1)
	$q = 4\pi \times 8.85 \times 10^{-12} \times 0.052 \times 3210$ $= 1.9 \times 10^{-8} \text{ C}$	(A1)

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Question	Answer	Marks
7(a)	output voltage / input voltage	M1
	input (voltage) is difference between (inverting and non-inverting) inputs	A1
7(b)	- reduces the gain - greater bandwidth - more stable <i>Any two points, 1 mark each</i>	B2
7(c)(i)	inverting amplifier	B1
7(c)(ii)	X marked anywhere between right-hand edge of $480\ \Omega$ resistor, left-hand edge of $1.2\ \text{k}\Omega$ resistor and the inverting input	B1
7(c)(iii)	gain = $(-R_f / R_i)$	C1
	= $(-1200 / 480)$	A1
	= -2.5	
7(c)(iv)	$V_{\text{IN}} = 6.5 / (-2.5)$ $= -2.6\ \text{V}$	A1
7(c)(v)	$(-2.5) \times (-5.4) = +13.5\ \text{V}$, and so output saturates $V_{\text{OUT}} = (+)8.0\ \text{V}$	A1

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Question	Answer	Marks
8(a)(i)	arrow from Q pointing downwards, labelled B	B1
8(a)(ii)	arrow from Q pointing towards P, labelled F	B1
8(b)(i)	force is proportional to product of both currents (I and $2I$) or Newton's third law	B1
	forces are equal	B1
8(b)(ii)	opposite	B1

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Question	Answer	Marks
9(a)(i)	emission of electrons (from a metal surface)	B1
	when electromagnetic radiation is incident (on electrons)	B1
9a(ii)	<u>minimum</u> energy required for an electron to leave surface	B1
9(b)(i)	threshold (frequency)	B1
9(b)(ii)	- photons are (discrete) packets of energy - energy of photons depends on frequency (of EM radiation) - electrons can only absorb a single photon (of energy) <i>Any two points, 1 mark each</i>	B2
	emission only possible if photon energy is at least the work function	B1
9(b)(iii)	work function = $hf_0 = 6.63 \times 10^{-34} \times 6.93 \times 10^{14}$	C1
	$= 4.59 \times 10^{-19} \text{ (J)}$ $= 4.59 \times 10^{-19} / 1.60 \times 10^{-19} \text{ (eV)}$ $= 2.87 \text{ eV}$	A1

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Question	Answer	Marks
10(a)(i)	to increase the magnetic flux linkage (between the coils)	B1
10(a)(ii)	to reduce energy losses	B1
	by reducing induced currents	B1
10(b)(i)	maximum $V_{\text{OUT}} = 12\,000 \times (625 / 25\,000)$ $= 300\text{ V}$	A1
10(b)(ii)	r.m.s. current $= 300 / (640 \times \sqrt{2})$ $= 0.33\text{ A}$	A1
10(b)(iii)	sketch: sinusoidal shape in positive half of the graph, sitting with 'minima' resting on the time-axis (at $P = 0$)	B1
	each 'cycle' shown repeating every 20 ms	B1
	maximum P shown as 140 W	B1
10(c)	power curve is symmetrical about the midpoint (on the power axis)	B1
	mean power is half the peak power	B1

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Question	Answer	Marks
11(a)	generates <u>ultrasound</u>	B1
	detects <u>reflected</u> ultrasound	B1
	applied p.d. causes crystal to vibrate or vibrations cause crystal to generate an e.m.f.	B1
11(b)(i)	product of density and speed	M1
	speed of ultrasound in medium	A1
11(b)(ii)	difference between (the specific acoustic impedances)	C1
	• if similar/same then reflection coefficient is zero/very low • if very different then reflection coefficient is (nearly) 1 • the lower the difference means lower the reflection coefficient (<i>any one point</i>)	A1

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Question	Answer	Marks
12(a)(i)	cannot predict when a particular nucleus will decay or cannot predict which nucleus will decay next	B1
12(a)(ii)	(decay is) not affected by external (environmental) factors	B1
12(b)(i)	$A = A_0 \exp(-\lambda t)$ and so $\ln A = \ln A_0 - \lambda t$ gradient of line = $(-)\lambda$	C1
	$\lambda = (36.4 - 35.0) / (20 - 0)$ $(= 0.07(0) \text{ min}^{-1})$	C1
	half-life = $\ln 2 / \lambda$ $= \ln 2 / 0.070$ $= 10 \text{ min}$	A1
	or	
	$A_0 = \exp(-36.4) = 6.43 \times 10^{15} \text{ (Bq)}$	(C1)
	$A_0 / 2 = 3.21 \times 10^{15} \text{ (Bq)}$, so $\ln(A_0 / 2) = 35.7$	(C1)
	read off half-life = 10 min	(A1)
	or	
	(at one half-life,) $\ln A = 36.4 - \ln 2$	(C1)
	$= 35.7$	(C1)
	read off half-life = 10 min	(A1)

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Question	Answer	Marks
12(b)(ii)	$A = \lambda N$	C1
	$N = \text{mass} / (\text{nucleon number} \times u)$ or $N = (\text{mass} / \text{nucleon number}) \times N_A$	C1
	$\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-7}) / (\text{nucleon number} \times 1.66 \times 10^{-27})$ or $\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-4} \times 6.02 \times 10^{23}) / \text{nucleon number}$ nucleon number = 62	A1